Feb 16, 2026 in Energy & Infrastructure

The 35 Gigawatt Gap

Data center electricity consumption is projected to double by 2030. This analysis examines regional supply constraints across the US, China, and Europe, and the emerging role of nuclear power in addressing them.

945 TWh	35 GW	2,600 GW	~10 GW	$300B+
Projected data center demand, 2030	US energy gap by 2030	US grid interconnection queue	Big Tech nuclear commitments	Big Tech AI capex, 2024-2026
IEA	Bloom Energy	LBL	Various	WSJ

In early 2025, PJM Interconnection, the grid operator for 13 US states and **65 million people**, revealed that its interconnection queue had swelled to **over 2,600 GW** of pending requests, more than twice the total installed capacity of the US power grid. A large share of that queue is data centers.

By 2030, the world's data centers will consume as much electricity as Japan. In some US regions, summer blackout risk is already classified as **"elevated"** by the grid reliability authority, and the surge in AI-driven power demand hasn't even peaked yet.

These are the findings from the IEA's **Energy and AI** report, published in January 2025. According to the IEA, global data center power consumption is on track to reach **945 TWh by 2030**, more than doubling from 415 TWh in 2024. A single AI query consumes **around 10x the electricity** of a conventional search. As AI scales from niche tool to ubiquitous infrastructure, the electricity bill is scaling with it.

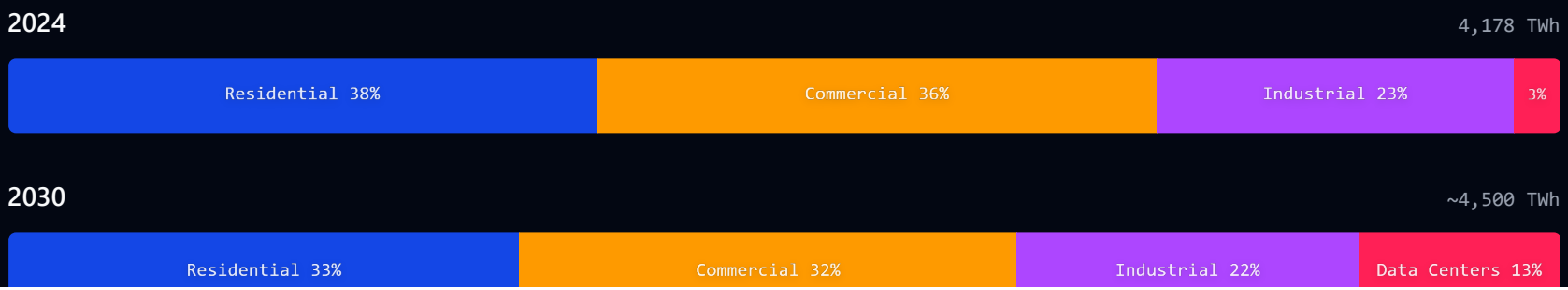
The question is not whether demand will grow. That trajectory is locked in by hundreds of billions in committed capital expenditure. The question is whether the electricity supply can keep up.

TL;DR

- The US and China account for 80% of the growth. Only 20% of projects in the US interconnection queue ever reach operation, with a 5-year median wait.
- Every major tech company is now investing in nuclear power, but most of that capacity won't arrive until the 2030s.
- By **2027-2028**, demand will meaningfully outpace committed supply, putting upward pressure on electricity prices and creating competition between data centers and households for grid capacity.

Where Does US Electricity Go?

Share of total US electricity consumption by sector. Data centers grow from a sliver to a significant block.





Sources: EIA Electric Power Monthly (2024), IEA Energy and AI (2025 projections). Data center share includes hyperscale and enterprise facilities.

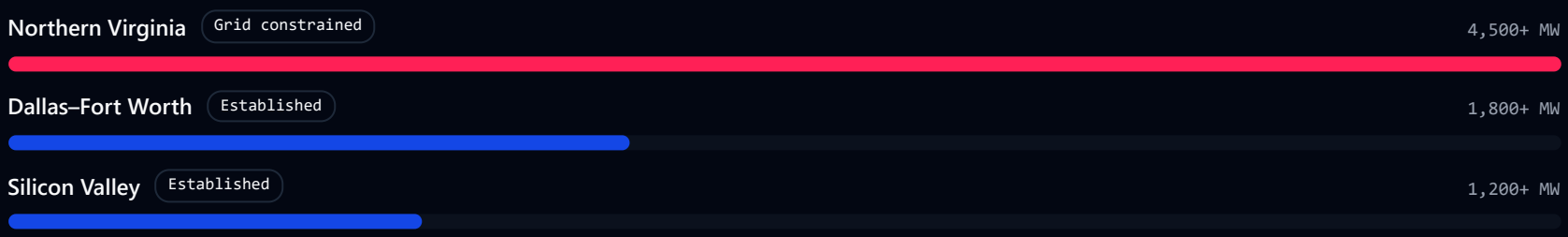
Where the Demand Is Growing

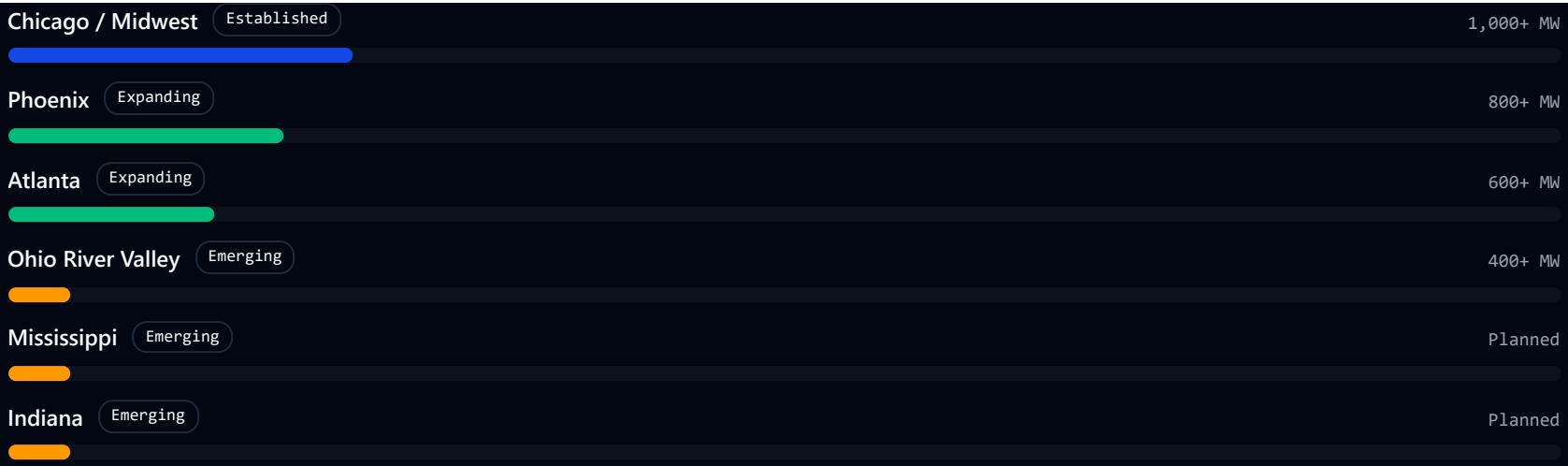
Data center electricity consumption is not evenly distributed. The United States dominates with approximately 45% of global demand, followed by China at 25% and Europe at 15%. These three regions account for 85% of all data center electricity consumption, and nearly all of the projected growth.

The trajectories differ markedly by region. The US is adding the most absolute capacity, driven by hyperscaler buildouts from Microsoft, Amazon, Google, and Meta concentrated in Virginia, Texas, and the Midwest. China is growing faster in percentage terms (+170% by 2030), fueled by domestic AI competition and state-backed data center expansion. Europe grows more modestly (+70%), constrained by stricter energy regulations and limited grid capacity.

US Data Center Hubs by Capacity

Grid constraints in established hubs are pushing new builds to the Midwest and Southeast.



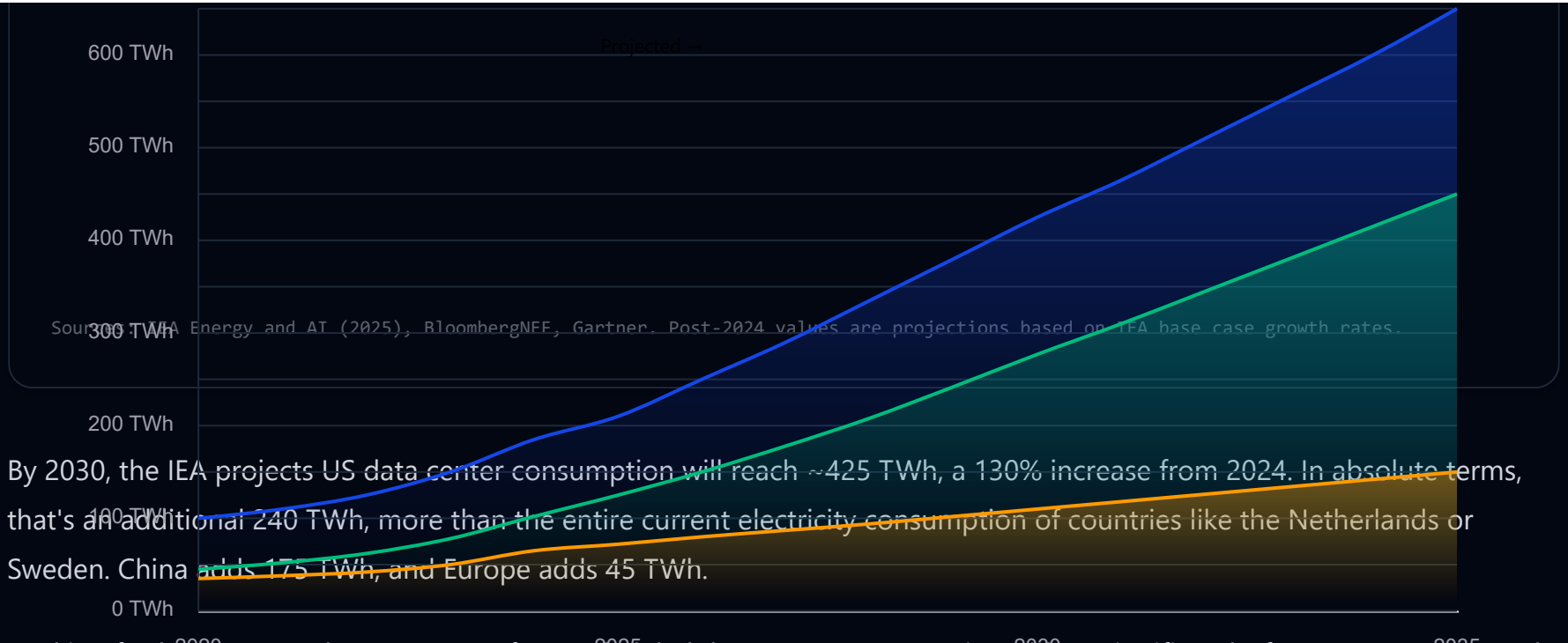


Sources: Cushman & Wakefield Data Center Market Report (2025), CBRE North America Data Center Figures. Capacity figures are approximate operational IT load.

Data Center Electricity Demand by Region (TWh)

Historical 2020-2024 and projected 2025-2035. Dashed line marks projection boundary.





By 2030, the IEA projects US data center consumption will reach ~425 TWh, a 130% increase from 2024. In absolute terms, that's an additional 240 TWh, more than the entire current electricity consumption of countries like the Netherlands or Sweden. China adds 175 TWh, and Europe adds 45 TWh.

Looking further to 2035, the IEA's range for *total global data center consumption* widens significantly, from 700 to 1,700 TWh, depending on the pace of AI adoption and efficiency gains. BloombergNEF projects US data center power demand alone could reach **106 GW by 2035**, nearly tripling current levels.

The Supply Crisis

Growing demand would be manageable if supply could keep pace. So far, it isn't.

Nowhere is this more visible than Loudoun County, Virginia, home to the densest cluster of data centers on Earth. The county hosts **roughly 200 facilities** drawing over 5 GW of power, with another 100+ in development. An estimated **70% of the world's internet traffic** is routed through Loudoun's "Data Center Alley," and the sector now accounts for **roughly 42% of the county's tax revenue**. Residential electricity bills have risen as **Dominion Energy requests rate increases** to fund grid

upgrades driven by data center load. Local officials now require developers to fund substation expansions as conditions for **zoning approval**. Substations built to serve suburban homes are now being sized for industrial loads, and residents are pushing back.

2,600 GW
Capacity waiting in the US grid interconnection queue
LBL / DOE

~5 years
Median wait from application to commercial operation
LBL / DOE

80%
Of queued projects that withdraw before reaching operation
LBL / DOE

The US grid interconnection queue tells the story. As of 2025, **2,600 GW** of generation capacity, more than twice the entire installed capacity of the US power plant fleet (~1,280 GW), is waiting for grid connection. For data centers specifically, some projects face delays of up to **12 years**.

> *"The North American Electric Reliability Corporation (NERC) has warned of "elevated risk" of summer electricity shortfalls starting in 2026 across all three US grid regions. The IEA estimates that up to 20% of planned data center projects could face delays without significant investment in transmission infrastructure."*

Bloom Energy's research puts it starkly: data centers face a **35 GW energy gap by 2030**. In response, approximately 30% of data center sites are expected to deploy onsite power generation as a primary energy source, a workaround that wouldn't be necessary if the grid could deliver.

The problem isn't a lack of proposed generation; it's the time it takes to build and connect it. A new natural gas plant takes 3-5 years. Utility-scale solar and wind take 2-4 years, but face interconnection delays. A conventional nuclear plant takes 10-15 years. Even the fastest new capacity additions can't match the pace at which hyperscalers are breaking ground on new data centers.

Big Tech Goes Nuclear

Facing grid constraints and multi-year interconnection queues, the largest technology companies have turned to an unlikely solution: nuclear power. What started as a single headline-grabbing deal in 2024 has become an industry-wide strategy.

The logic is straightforward. Nuclear provides baseload power: reliable, 24/7 electricity with no carbon emissions, at a density that renewables can't match. A single nuclear reactor can power a major data center campus on a relatively small footprint. And unlike grid-connected renewables, behind-the-meter nuclear sidesteps the interconnection queue entirely.

THE WASTE QUESTION

90,000 metric tons of spent fuel sit in temporary US storage across 75+ sites. No country operates a permanent repository (Finland's Onkalo is first, 2025).

A 2022 Stanford/PNAS study found SMRs may produce **2-30x more waste** per unit of energy than conventional reactors. The industry disputes the magnitude.

Safety is statistically strong (0.03 deaths/TWh, on par with wind and solar), yet perceived risk drove Germany to shut its last reactors in 2023.

Sources: GAO, WNA, PNAS, Our World in Data

How Much Is 1 GW of Power?

One gigawatt, enough to power a major city. Here's what it takes to produce it, and what it can run.

What does it take to produce 1 GW?

Each ■ = 1 unit (reactor, turbine) except solar (each ■ ≈ 10,000 panels). Nuclear first, then increasing.

Nuclear 1 units · 92% CF

1 reactor (~1 GW)

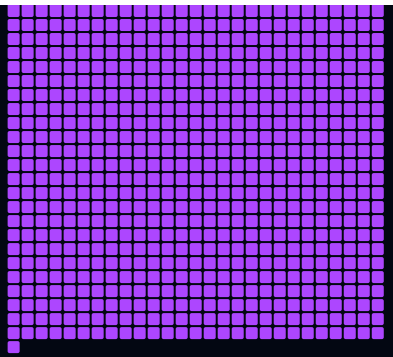
Gas 3 units · 87% CF

3 combined-cycle gas turbines

Wind ~300 units · 35% CF

Solar ~10M units · 25% CF

~300 onshore turbines (3–5 MW each)



~10 million panels (each ■ ≈ 10k)

Sources: DOE, Nuclear Energy Institute, EIA. Unit counts account for capacity factor, i.e. the actual equipment needed to deliver 1 GW continuously.

What can 1 GW power?

Each ■ = 1 data center or 1 flight. For households, each ■ ≈ 1,000 homes.

Data Centers ~10



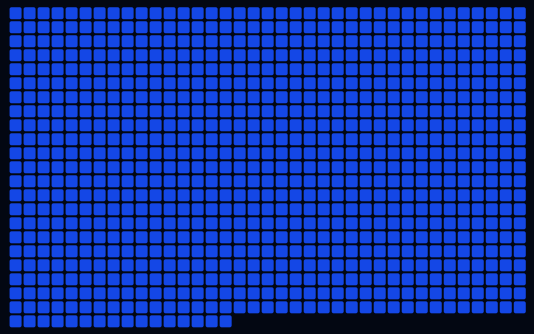
~10 hyperscale campuses (~100 MW each)

NYC → London ~30 / day



~30 transatlantic 747 flights (energy eq.)

Households ~830k



~830,000 US homes (each ■ ≈ 1k homes)

Sources: EIA (household avg 10,500 kWh/yr = 1.2 kW continuous load), Uptime Institute / LBNL (hyperscale campus ~100 MW avg), FlightDeckFriend / Wikipedia (747 burns ~80k L jet fuel per transatlantic flight, jet fuel 9.6 kWh/L).

5	7	~10 GW
Companies with nuclear commitments	Announced nuclear deals	Total committed nuclear capacity

Microsoft | Amazon | Google | Meta | Oracle

Microsoft

835 MW

Reactor restart

TOTAL CAPACITY	INVESTMENT	DEALS
835 MW	$1B+ (DOE loan)	1

20-year PPA with Constellation Energy to restart Three Mile Island Unit 1 (the undamaged reactor at the site of America's worst nuclear accident). Renamed the Crane Clean Energy Center. DOE closed a $1B loan guarantee in Nov 2025.

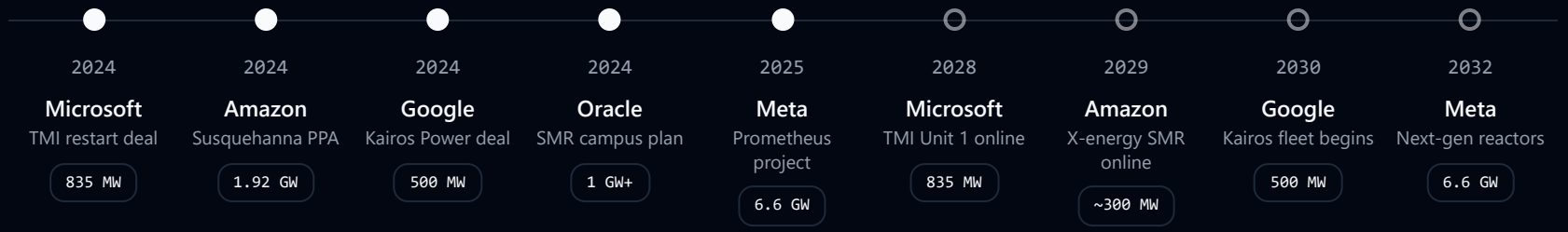
Constellation Energy | 835 MW | TMI Unit 1 restart | 2028

Source: Financial Content

Big Tech nuclear power commitments as of February 2026. Capacity figures represent announced targets, not operational capacity.

Nuclear Deal Timeline

When each deal was announced (solid) and when reactors target operation (outline). Most capacity arrives after 2030, but the energy gap opens in 2027.



Sources: Company announcements, DOE filings, Fortune, Financial Content. Target dates are company-stated and subject to regulatory approval.

The Reality Check

Collectively, these commitments represent approximately **8-10 GW** of nuclear capacity. That is meaningful, but a fraction of the 50+ GW of new capacity the DOE says is needed from data centers alone by 2030. Most small modular reactor (SMR) technology remains unproven at commercial scale. NuScale, the only SMR vendor with full NRC certification, saw its first commercial project **cancelled in 2023** after costs ballooned from $5.3B to $9.3B. No commercial SMR operates in the Western world yet.

The near-term supply will come from existing reactor PPAs (Microsoft, Amazon, Meta) and gas plants, not SMRs. The nuclear bet is a 2030s play, not a 2020s solution.

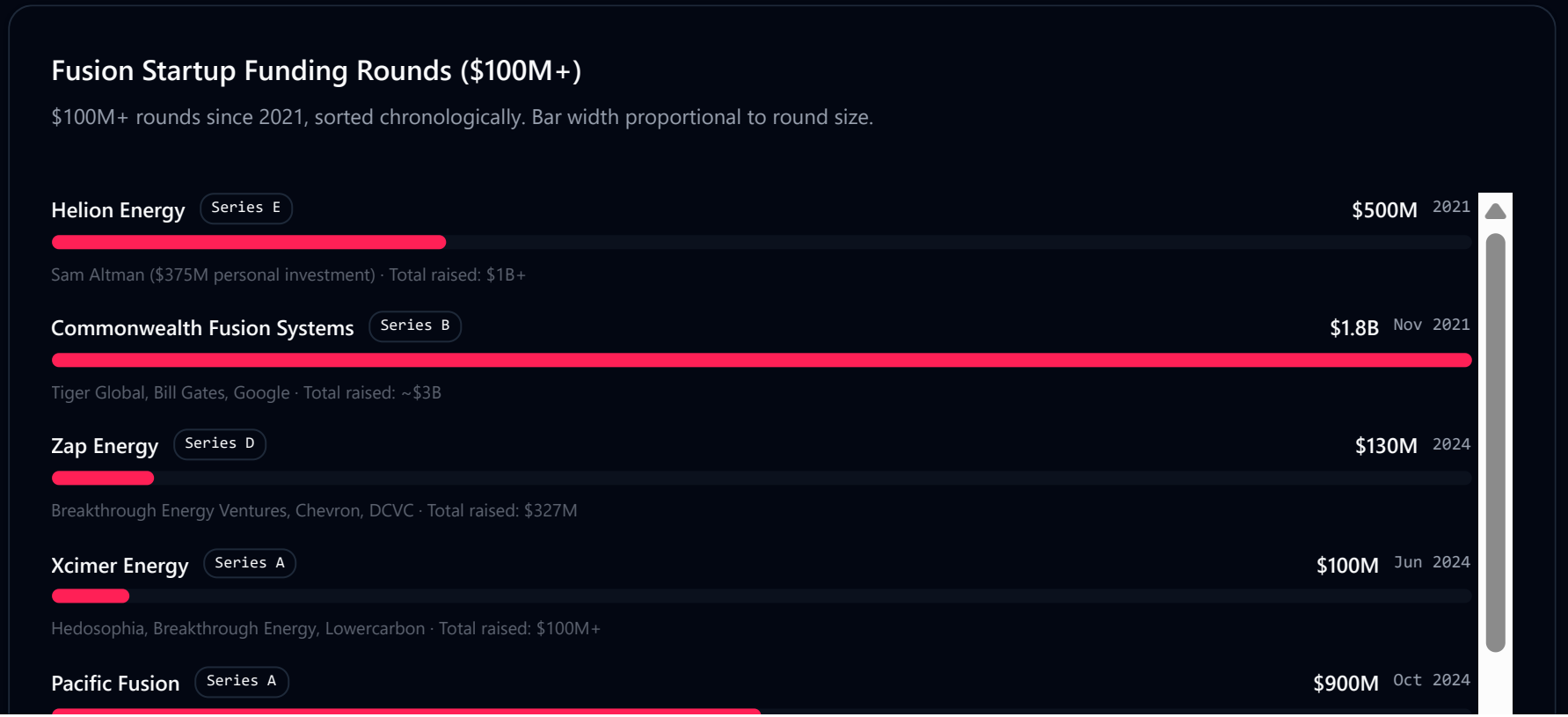
Beyond Fission: The Fusion Bets

Some hyperscalers are placing even longer-horizon bets on fusion. Microsoft signed the first-ever commercial fusion power purchase agreement, 50 MW from **Helion Energy**, targeting 2028. Helion broke ground on its facility in Malaga, Washington in August 2025. Google followed with a **200 MW agreement** with Commonwealth Fusion Systems in mid-2025, alongside a longstanding partnership with TAE Technologies dating back to 2014.

These commitments are tiny relative to the fission deals (hundreds of megawatts, not gigawatts), and no fusion reactor has yet produced net electricity commercially. But they signal that Big Tech is willing to fund the full spectrum of nuclear technology, from 1950s reactor restarts to physics that hasn't been proven at scale.

In fact, the broader fusion investment landscape tells a striking story. According to the **Fusion Industry Association**, cumulative private and public funding in fusion companies has reached roughly $15 billion as of mid-2025, with $2.64 billion raised in the twelve months to July 2025 alone, a 178% year-on-year increase. The US accounts for over half of global fusion funding (~$8 billion across 42 companies), with China second at ~$5 billion.

Major Rounds | Power Agreements | Industry Totals



General Catalyst, Eric Schmidt, Patrick Collison, Ken Griffin, BEV · Total raised: $900M

Tokamak Energy – **$125M** Nov 2024

In-Q-Tel, Hans-Peter Wild · Total raised: $336M

Helion Energy Series F **$425M** Jan 2025

Sources: FIA 2025 Report, TechCrunch, company announcements. CFS alone accounts for ~$3B, roughly one-third of all private fusion capital globally.

Fusion energy investment as of February 2026. No fusion reactor has yet produced net electricity commercially.

The Gap: Demand vs. Supply

Bringing the demand and supply trajectories together reveals the core problem. US data center power demand is projected to grow from around 28 GW today to 106 GW by 2035. But committed new supply, including renewables, gas, and nuclear, is not keeping pace.

US Data Center Power: Demand vs. Committed Supply (GW)

The gap between the two lines represents unfilled capacity. Supply = renewables + gas + nuclear committed or under construction.





The chart tells a ~~clear story. Through 2025, existing supply approximately matches demand. But starting in~~ **2026–2027**, a gap opens and widens rapidly. By 2030, projected demand reaches 75 GW while committed supply stands at 58 GW, a **17 GW shortfall**. Bloom Energy's independent analysis arrives at an even larger estimate of **35 GW**, using a broader definition of demand that includes edge computing and enterprise AI workloads beyond hyperscale data centers.

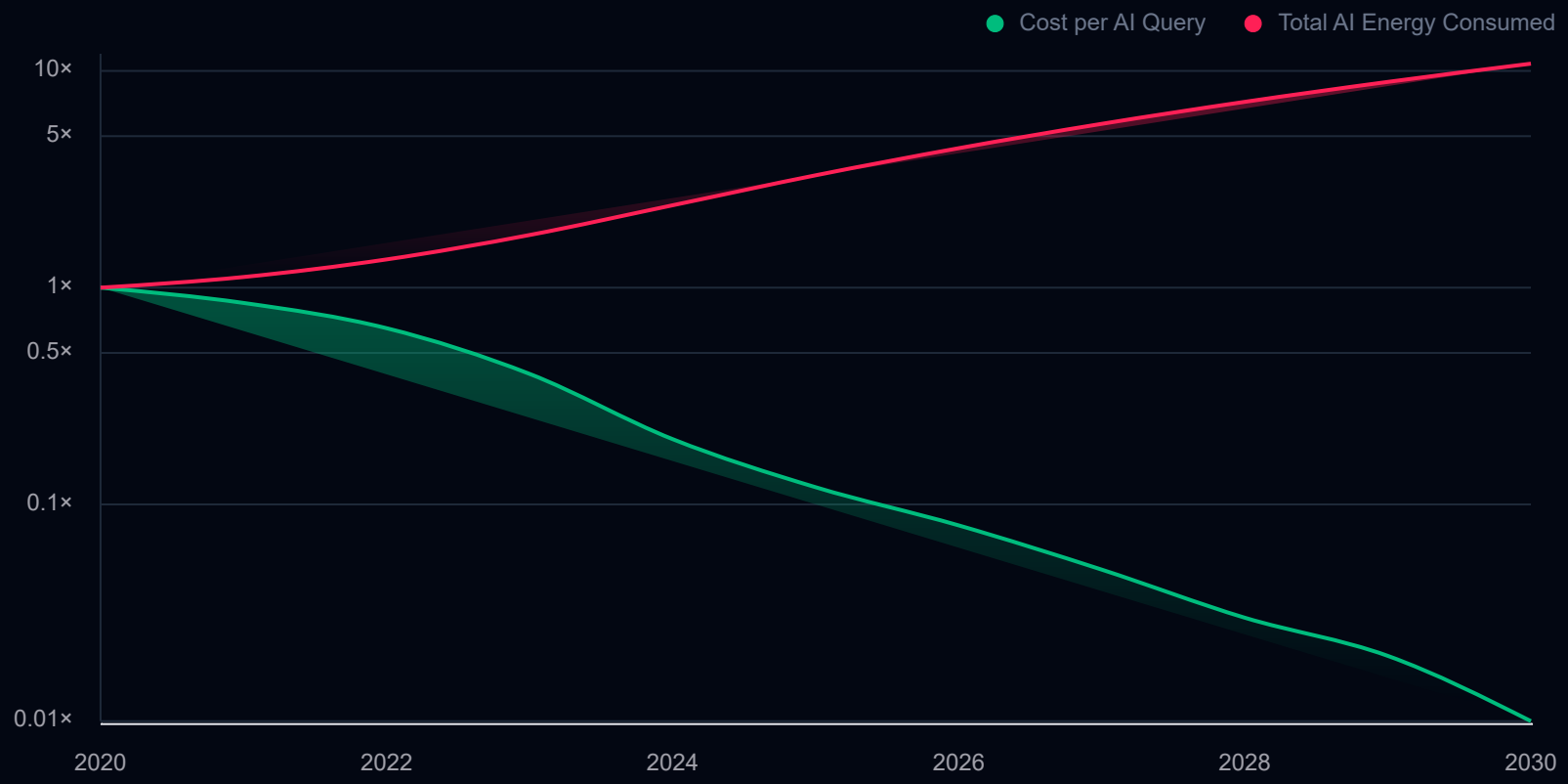
The gap peaks in the early 2030s before beginning to narrow as longer-lead-time projects, particularly nuclear SMRs and large renewable installations, come online. But even by 2035, committed supply falls short of projected demand by approximately 14 GW.

> ***"The critical period is 2027-2032. Demand is growing at its fastest rate, driven by AI buildouts already under construction. But the corresponding supply takes 3-7 years to come online. The decisions that would close the 2030 gap needed to be made in 2025-2026, and many of them weren't."***

The Efficiency Mirage

The Jevons Paradox: Cheaper Queries, More Total Energy

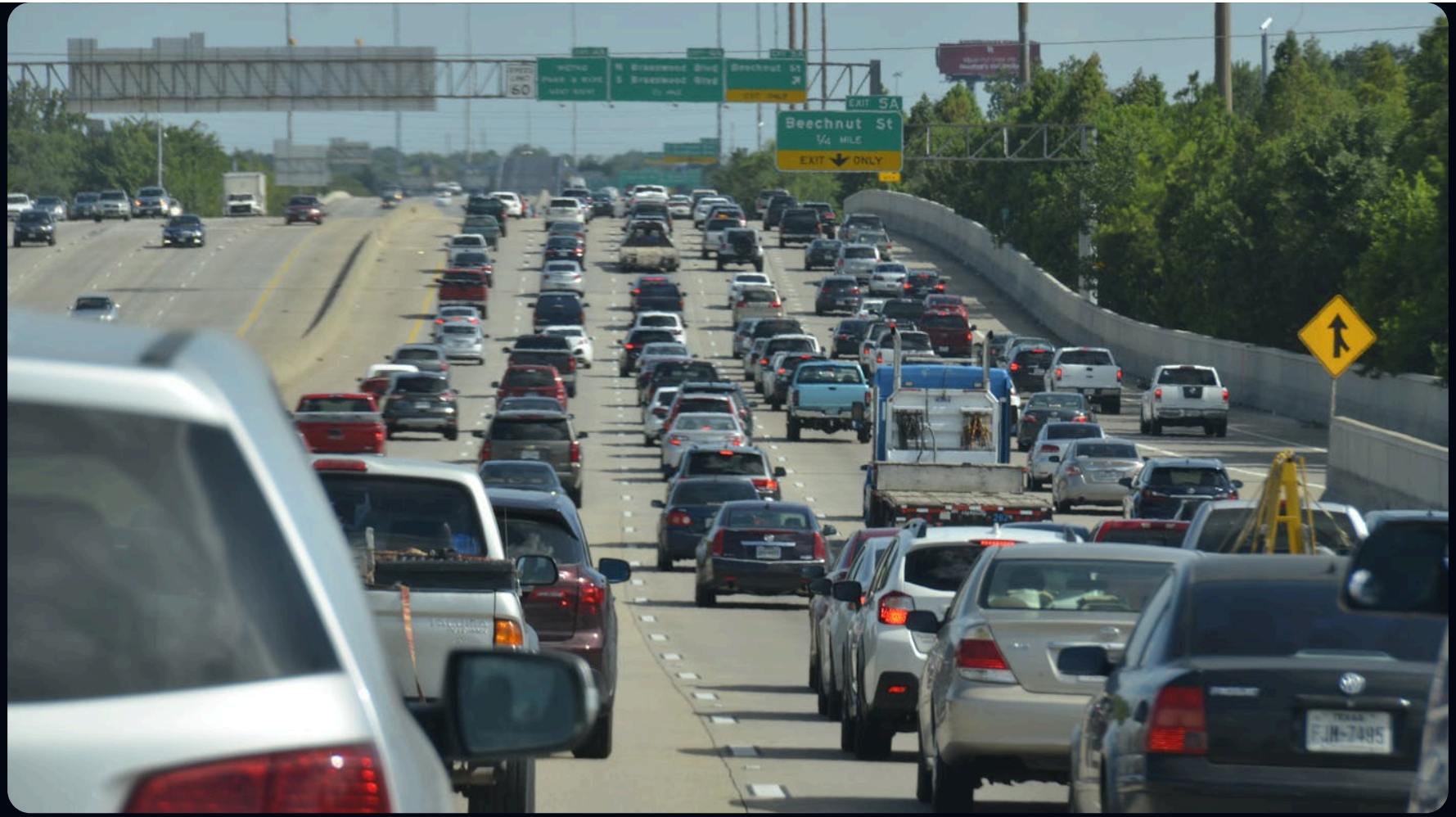
Both series relative to 2020 (= 1×). Log scale. Cost per query has dropped ~100×, yet total energy consumed has risen ~10×.



Illustrative model. Cost trajectory based on OpenAI API pricing trends and hardware efficiency gains (TPU, Trainium, Blackwell). Energy trajectory based on IEA demand projections. Log scale, 2020 = 1×.

A natural counterargument: won't AI become more efficient? It is true that inference costs are falling rapidly. Model distillation, quantization, and purpose-built silicon (Google's TPUs, Amazon's Trainium, Nvidia's Blackwell architecture) are all driving down the energy cost per query. OpenAI's pricing for GPT-4-class models has dropped by an order of magnitude since 2023.

But historically, efficiency gains in computing have not reduced total energy consumption. This pattern is known as **Jevons paradox**: when a resource becomes cheaper to use, new applications become viable, and total usage tends to rise. If cheaper inference leads to AI being embedded in more products and workflows, aggregate power demand could grow even as per-query costs fall. The IEA's wide forecast range (700-1,700 TWh by 2035) reflects this uncertainty.



Induced demand in action: more lanes, same gridlock. The same logic applies to AI efficiency — cheaper inference drives more usage, not less total energy.

Photo by Luigi Alvarez on Pexels. Free to use.

Three Grids, Three Strategies

The US, China, and Europe face the same problem: surging data center demand hitting grid limits. But their responses differ sharply, reflecting different political systems as much as different energy policies.

China: State-Directed Buildout

China's approach is infrastructure-first. The **"East Data, West Computing"** initiative, launched in 2022, is constructing eight national computing hubs and ten data center clusters, channeling compute demand from coastal megacities to resource-rich western provinces. By mid-2024, over **$6 billion had been directly invested** in these hubs, driving total investment exceeding $28 billion.

The backbone: ultra-high-voltage (UHV) transmission. China has built over 50,000 km of UHV lines and is investing **$200 billion+ in transmission infrastructure**. The Gansu–Zhejiang 800kV DC line, a **$5 billion project** commissioning in 2026, will transmit 4 GW of renewable energy from the interior to the coast. In early 2025, China launched the Future Network Test Facility, a **55,000 km distributed AI computing network** spanning 40 cities, achieving 98% of single-center efficiency.

Powering these centers demands energy at an unprecedented scale. In the Kubuqi Desert in Inner Mongolia, China is building what NASA calls a **"Solar Great Wall"**: a 400 km long, 5 km wide solar installation targeting 100 GW of capacity by 2030. Over 5 GW is already installed. The satellite imagery tells the story:



December 2017: barren desert



December 2024: solar panels covering the desert

NASA Earth Observatory images by Michala Garrison, using Landsat data from USGS. Public domain.

The strategic implication is significant. China's centralized planning sidesteps the interconnection queue bottleneck that plagues the US. When **US AI experts visited China in 2025**, they returned, in Fortune's words, "stunned" by the infrastructure advantage. **Goldman Sachs estimates** Chinese data center demand could reach 600 TWh by 2030, and unlike in the US, supply may actually keep pace.

Europe: Regulation First

Europe's approach is constraint-based. Ireland, home to a disproportionate share of European cloud infrastructure due to favorable tax policy and Atlantic cable landings, **imposed a moratorium** on new data center grid connections in 2021 after the sector threatened to consume 30% of national electricity. Amsterdam enacted similar restrictions.

Ireland **lifted the moratorium in late 2025**, but with strings: new facilities must now source **80% of their energy from on-site renewables** or battery storage. The European Commission is launching **new measures to limit data center energy consumption** across the EU in 2026.

The risk for Europe is asymmetric. By constraining data center supply while the US and China race to build, Europe may preserve grid stability at the cost of AI sovereignty. The modest growth trajectory in the regional demand chart (from 65 TWh to 150 TWh by 2035) reflects not just limited grid capacity, but a deliberate policy choice that may leave the continent dependent on American and Chinese AI infrastructure.

What This Means

Electricity Prices

Data center hubs (Northern Virginia, central Texas, Ohio River Valley) will see sustained upward pressure on electricity prices. Dominion Energy has already requested rate increases to fund data-center-driven grid upgrades. Residential ratepayers will absorb part of the cost.

Grid upgrades for data centers are increasingly reflected in residential electricity rates.

Data Center Geography

Northern Virginia, with 300+ facilities and the densest cluster on Earth, is saturated. New builds are shifting to Mississippi, Indiana, and Wisconsin, where grid capacity still exists.

Grid capacity is becoming the primary constraint on where new AI infrastructure gets built.

Grid Reliability

NERC warns of elevated summer shortfall risk starting 2026. Adding tens of GW of data center load to grids sized for slow residential growth creates real blackout risk during extreme heat events.

In regions with constrained grids, data center load and residential demand increasingly draw from the same limited

Nuclear Energy

Big Tech's nuclear commitments are the largest private-sector bet on nuclear in decades. If even a fraction of SMR projects succeed, it reshapes the economics of nuclear. But NuScale's first project ballooned from $5.3B to $9.3B before cancellation.

supply.

These are the largest private-sector nuclear commitments in decades, though most capacity won't arrive until the 2030s.

Geopolitical Competition

China's centralized grid planning and $200B+ transmission investment may give it an infrastructure edge. Europe's regulatory approach risks ceding AI sovereignty. The US interconnection queue (2,600 GW stalled) is becoming a strategic liability.

Energy infrastructure is emerging as a key factor in geopolitical AI competition.

Could AI deployment actually slow down? It's possible. If electricity costs rise significantly and grid connections remain backlogged, some AI deployments will be delayed or relocated offshore. But the competitive dynamics of AI, where falling behind in capability means losing market position, make voluntary slowdowns unlikely. More probable: companies will pay whatever it takes for power, and the costs will flow downstream to consumers and ratepayers.

The bottom line: the electricity infrastructure that powered the internet era was built over decades. The AI era is attempting to consume a comparable amount of power in years. China is building its way out of the problem with state-directed megaprojects. Europe is regulating its way around it. The US sits between market-driven demand and slow-moving supply infrastructure. How each region resolves that tension will shape who has the capacity to run the next generation of AI, and who doesn't.

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